



HELIOS

One Operational World

A concise guide to AI-assisted and autonomous Site operations

AI + AUTONOMY + PEOPLE + ROBOTICS

Proposed user experience based on the currently understood Helios environment

Version 0.1 – August 2026

What Helios actually is

One governed environment for understanding conditions, coordinating response, supervising autonomy, and proving outcomes.

Operational world

Digital Twin, Sites, zones, routes, sensors, robotic assets, human teams, evidence, and readiness.

Governed action

Attention, Incidents, Missions, resources, Commands, intervention, observed effects, and Records.

Cortex intelligence

Assessment, recommendations, policy-bounded intent, memory, learning, and Agent coordination.

The principle

Existing systems retain the authority they should own. Helios connects them into one operational and autonomy layer.

One Site. One event. One shared context.

02:14 – west restricted perimeter

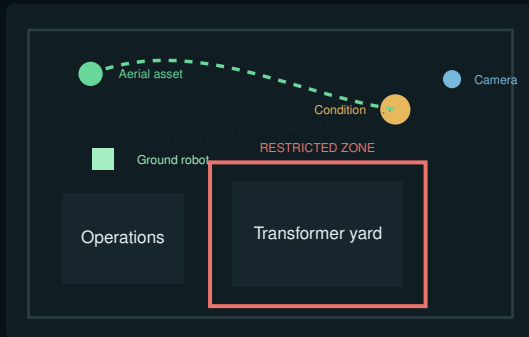
- ▶ Fixed visible camera reports motion.
- ▶ Thermal source is consistent with a person.
- ▶ No matching access event is present.
- ▶ Visual identity remains unknown.
- ▶ One aerial asset is available.
- ▶ A ground robot is staged nearby.

Attention created

Helios shows consequence, uncertainty, source freshness, nearby capabilities, route restrictions, reserve coverage, and current authority.

Selection is inspection. No map click, camera movement, or Cortex statement creates a physical effect.

NORTH RIVER ENERGY CAMPUS – FICTIONAL ILLUSTRATION



The same event at increasing authority levels

PHASE 1 – OPERATOR DIRECTED

Helios presents evidence, world context, qualified resources, route, readiness, and consequence. The operator creates and launches the Mission.

Human role: direct.

PHASE 2 – CORTEX ASSISTED

Cortex correlates evidence, explains uncertainty, compares options, and recommends an aerial Mission. The operator accepts, modifies, rejects, or asks why.

Human role: decide.

PHASE 3 – POLICY AUTONOMY

Evidence and policy qualify. Cortex initiates the approved observation Mission through normal readiness, resource, and Command admission.

Human role: supervise.

PHASE 4 – MULTI-AGENT COORDINATION

Site, Asset, Mission, and Operations Agents request and broker capabilities. Humans govern priorities, exceptions, policy, and consequence.

Conceptual operational view

CONCEPTUAL OPERATIONAL VIEW – PROPOSED USER EXPERIENCE

ATTENTION

WEST PERIMETER

Multi-source event
Age 00:21
Policy-initiated Mission

NORTH COVERAGE

Reserve reduced 8%

Human judgment not yet required

TACTICAL WORLD



Mission: Investigate west perimeter

CORTEX

Assessment
Person-like thermal source. Visual confirmation incomplete.

Policy
Restricted Perimeter Observation v4.

Why Helios acted
Two current sources, after-hours zone, route valid, reserve preserved.

MISSION AND RECEIPTS

Tasks: overhead view → track → inspect blind area → preserve evidence

Authorized **YES** Delivered **YES** Asset accepted **YES** Effect observation **PENDING**

HOLD REDIRECT RETURN ABORT

The Digital Twin is the operational world

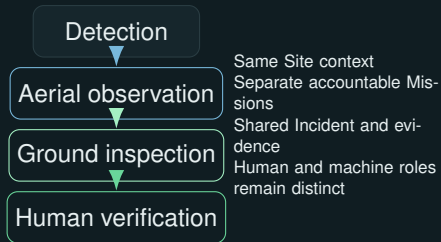
More than a 3D model

- ▶ Source geometry, frames, calibration, and uncertainty.
- ▶ Buildings, zones, routes, POIs, docks, and recovery.
- ▶ Cameras, sensors, coverage, and freshness.
- ▶ Asset deployments, capability, and readiness.
- ▶ Missions, policy, evidence, and revision.

Cortex uses structured context

Location, reachability, coverage, restrictions, resource qualification, route consequence, and evidence relationships.

One accountable Mission picture



Trust is visible

Authorized → Delivered → Accepted → Observed → Outcome known

Cortex manages complexity. Attention protects focus.

ATTENTION answers one question

What actually needs me now?

New uncertainty, autonomous activity, degraded communications, resource conflict, unknown Command result, stale evidence, policy exception, or decision due.

Major Cortex roles

Site Agent Asset Agent Mission Agent
Operations Agent

Specific accountable roles under one user-facing Cortex identity.

OPERATIONAL EXERCISES

The same world, Missions, assets, Cortex, Attention, controls, and Records can be exercised under unmistakable simulation authority.

- ▶ Shift and Mission practice.
- ▶ Communications and sensor degradation.
- ▶ Autonomous-policy evaluation.
- ▶ Operator intervention and After-Action Review.

Learning remains governed

Evidence → candidate → simulation or replay → review → approved version → monitoring and rollback.

From one Site to the enterprise

North RiverReady – 2 Missions

East Switch

Enterprise Operations + Cortexposture, Atten

Cross-Site requestaerial thermal capability

Hillcrest HydroLocal operation

The payoff

Your systems. Your Sites. One operational world.

See the operation. Coordinate the response. Prove the outcome.